

GROUP THEORY

Non-residually finite central extensions of arithmetic groups. Note (*) by **Pierre Deligne**,
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A variant of the known methods for treating the congruence subgroup problem gives results such as the following. Let $\widetilde{\mathrm{Sp}}(2n, \mathbf{Z})$ be the inverse image of $\mathrm{Sp}(2n, \mathbf{Z})$ in the universal covering of $\mathrm{Sp}(2n, \mathbf{R})$; this is a central extension of $\mathrm{Sp}(2n, \mathbf{Z})$ by

$$\pi_1 \mathrm{Sp}(2n, \mathbf{R}) = \mathbf{Z}.$$

If $n \geq 2$, then $\widetilde{\mathrm{Sp}}(2n, \mathbf{Z})$ is not separated for the topology of subgroups of finite index. More precisely, every subgroup of finite index in $\widetilde{\mathrm{Sp}}(2n, \mathbf{Z})$ contains $2\pi_1 \mathrm{Sp}(2n, \mathbf{R})$.

1. S -arithmetic groups. Let G be an almost simple simply connected isotropic group over a global field F . To simplify the statements, we shall also assume that $G(F)$ is perfect, that is, equal to its commutator subgroup; this is the case except possibly when G is a trialitarian form of D_4 . Let S be a finite set of places of F , containing all the archimedean places, and suppose that

$$(i) \quad \sum_{v \in S} \mathrm{rank}(G/F_v) \geq 2.$$

For each place v of F , put $G_v = G(F_v)$. For each finite set T of places, put similarly

$$G_T = \prod_{v \in T} G_v, \quad G_{T'} = \prod_{v \notin T} G_v$$

where the second product is restricted. Thus, for $\mathbf{A}$ the adèle ring of F ,

$$G(\mathbf{A}) = G_T \times G_{T'}.$$

A subgroup of $G(F)$ is called S -congruence if it is the inverse image of a compact open subgroup of G_S . It is called S -arithmetic if it is commensurable with an S -congruence subgroup, equivalently if it contains, with finite index, a subgroup of finite index in an S -congruence subgroup. The S -congruence subgroups, respectively the S -arithmetic subgroups, form a fundamental system of neighborhoods of the identity for group topologies $\mathcal{T}(c)$ and $\mathcal{T}(a)$ on $G(F)$. For a fixed S -congruence subgroup Γ , one obtains the same fundamental systems by considering only the S -congruence subgroups normal in Γ , respectively only the normal subgroups of finite index in Γ . This ensures the existence of the completions

$$G(F)^\wedge(c), \quad G(F)^\wedge(a)$$

of $G(F)$ for $\mathcal{T}(c)$ and $\mathcal{T}(a)$. The topology $\mathcal{T}(c)$ is the topology induced from G_S . Since $G(F)$ is dense in G_S by strong approximation, one has

$$G(F)^\wedge(c) = G_S.$$

The completion $G(F)^\wedge(a)$ is an extension of G_S by the profinite group

$$C(S, G) = \varprojlim \Gamma / \Delta,$$

where the inverse limit is over pairs $\Delta \subset \Gamma$ with Δ normal of finite index in Γ and Γ S -congruence.

For G of rank at least 2, Raghunathan ⁽¹⁾ proved that

$$(ii) \quad G(F)^\wedge(a) \text{ is a central extension of } G(F)^\wedge(c) = G_S,$$

and he finishes checking that the same conclusion holds under the sole hypothesis (i).

Let $\widetilde{G}_S$ be a topological central extension of G_S by a discrete group A , and let $\widetilde{G(F)}$ be the inverse image of $G(F)$ in $\widetilde{G}_S$. A subgroup of $\widetilde{G(F)}$ is called S -congruence if it is the inverse image of an S -congruence subgroup of $G(F)$, and is called S -arithmetic if it is commensurable with an S -congruence subgroup. We again denote the corresponding topologies by $\mathcal{T}(c)$ and $\mathcal{T}(a)$, and the separated completions by

$$\widetilde{G(F)}^\wedge(c), \quad \widetilde{G(F)}^\wedge(a).$$

Again $\widetilde{G(F)}^\wedge(c) = G_S$, while $\widetilde{G(F)}^\wedge(a)$ is an extension of G_S by a profinite group $\widetilde{C}(S, G)$, itself an extension of $C(S, G)$ by

$$\widehat{A} = \varprojlim A/(A \cap \Gamma) \quad (\Gamma \text{ } S\text{-arithmetic}).$$

Proposition 1. *If condition (ii) holds, then $\widetilde{G(F)}^\wedge(a)$ is a central extension of $\widetilde{G(F)}^\wedge(c) = G_S$.*

Recall that, if a group $\widetilde{H}$ is a central extension of a group H , then the commutator map $\widetilde{H} \times \widetilde{H} \rightarrow \widetilde{H}$ factors through a map, still denoted $(,) : H \times H \rightarrow \widetilde{H}$. In particular, since $\widetilde{G(F)}^\wedge(a)$ is a central extension of $G(F)^\wedge(a)$ by $\widehat{A}$, the commutator map

$$\widetilde{C}(S, G) \times \widetilde{G(F)}^\wedge(a) \rightarrow \widetilde{G(F)}^\wedge(a)$$

factors through $C(S, G) \times G(F)^\wedge(a)$. By (ii), it factors through a bimultiplicative map

$$C(S, G) \times G(F)^\wedge(a) \rightarrow \widehat{A}.$$

Since $G(F)$ is perfect and dense in $G(F)^\wedge(a)$, this map is trivial.

More concretely, the proposition says that, for every S -arithmetic subgroup Δ , the following hold.

(1a) For every conjugate Δ^g of Δ , with $g \in \widetilde{G(F)}$, there exists an S -congruence subgroup Γ such that

$$\Gamma \cap \Delta = \Gamma \cap \Delta^g.$$

(1b) $\widetilde{G(F)}$ acts trivially on the finite group

$$C_\Delta = \varprojlim \Gamma/(\Gamma \cap \Delta).$$

These conditions are also equivalent to

(1c) For every $g \in G(F)$, there exists an S -congruence subgroup Γ such that $(g, \Gamma) \subset \Delta$.

One has $\widetilde{C}(S, G) = \varprojlim C_\Delta$.

2. Central extensions of adelic groups. From now on we assume that

(iii) $\widetilde{G}_S$ is perfect.

The most interesting case would be that in which $\widetilde{G}_S$ is the universal central extension of G_S by a discrete group, the “universal covering” of G_S , but we do not want to assume the existence of such a universal extension.

A topological central extension E of $G(\mathbf{A})$ by a discrete group X is determined by the extensions E_S and $E_{S'}$ of G_S and $G_{S'}$ by X obtained by inverse image in E . Since G_S is perfect, one has

$$E \simeq (E_S \times E_{S'}) / \{(x, -x) \mid x \in X\}.$$

If the extension E_S is dominated by $\widetilde{G}_S$, that is, if it is obtained from $\widetilde{G}_S$ by pushout through a map $f : A \rightarrow X$, then f is unique, and giving E is the same as giving $E_{S'}$ together with f ; this is an equivalence of categories. In this case

$$E \simeq \widetilde{G}_S \times E_{S'} / \{(a, -f(a)) \mid a \in A\}.$$

Let $\mathcal{L}$ be the category of topological central extensions of $G(\mathbf{A})$ by a finite group, split above $G(F)$, and such that E_S is dominated by $\widetilde{G}_S$:

$$\begin{array}{ccccccc} 1 & \longrightarrow & X & \longrightarrow & E & \longrightarrow & G(\mathbf{A}) \longrightarrow 1 \\ & & & \searrow & \downarrow & & \\ & & & & G(F) & & \end{array}$$

It would amount to the same to require only that the extension be splittable over $G(F)$, since $G(F)$ is perfect and a splitting, if it exists, is unique. The category $\mathcal{L}$ is cofiltered. For (E, s) in $\mathcal{L}$, the closure in E of

$$(\text{image of } \widetilde{G}_S) \cdot sG(F)$$

maps to $G(\mathbf{A})$; it maps to a closed subset, because X is finite, and it contains $G_S G(F)$, which is dense by strong approximation. Thus the full subcategory L of $\mathcal{L}$ consisting of those (E, s) for which $(\text{image of } \widetilde{G}_S) \cdot sG(F)$ is dense in E is cofinal. Since there is at most one arrow between two objects of L , it is just a filtered ordered set.

Proposition 2. *Under hypotheses (ii) and (iii), there is a natural isomorphism between $\tilde{C}(S, G)$ and the inverse limit, over $\mathcal{L}$ or over L , of the groups X . Via this isomorphism, the natural map from A to $\tilde{C}(S, G)$ is the inverse limit of the maps $-f : A \rightarrow X$, where f is such that E_S is obtained from $\tilde{G}_S$ by pushout through f .*

Let E be a topological central extension of $G(\mathbf{A})$ by a discrete group X . To give a splitting s of E above $G(F)$ is to give an isomorphism between the extension of $G(F)$ by X obtained from E_S , and the opposite of the extension obtained from $E_{S'}$. If E_S is dominated by $\tilde{G}_S$ and f is as above, this is the same as giving a commutative diagram

$$\begin{array}{ccccccc} 1 & \longrightarrow & X & \longrightarrow & E_{S'} & \longrightarrow & G_{S'} \longrightarrow 1 \\ & & \uparrow -f & & \uparrow i & & \uparrow \\ 1 & \longrightarrow & A & \longrightarrow & \widetilde{G(F)} & \longrightarrow & G(F) \longrightarrow 1. \end{array}$$

It follows that $\mathcal{L}$ is equivalent to the category $\mathcal{L}'$ of commutative diagrams

$$(1) \quad \begin{array}{ccccccc} 1 & \longrightarrow & X & \longrightarrow & E_{S'} & \xrightarrow{q} & G_{S'} \longrightarrow 1 \\ & & & & \uparrow i & & \uparrow \\ & & & & \widetilde{G(F)} & \longrightarrow & G(F) \end{array}$$

where $E_{S'}$ is a topological central extension of $G_{S'}$ by a finite group. One reconstructs E as

$$\tilde{G}_S \times E_{S'} / \{(a, t(a)) \mid a \in A\},$$

and the splitting over $G(F)$ by

$$s(g) = (\tilde{g}, \tilde{t}\tilde{g}) \bmod (a, t(a)),$$

for $\tilde{g} \in \widetilde{G(F)} \subset \tilde{G}_S$ mapping to $g \in G(F)$. The subcategory L corresponds to the subcategory L' of diagrams (1) in which t has dense image.

Let $(E_{S'}, t)$ be in L' , and let $\mathcal{T}$ be the topology on $\widetilde{G(F)}$ induced by that of $E_{S'}$. If K is a compact open subgroup of $E_{S'}$ disjoint from X , then the intersections of $\Delta = t^{-1}K$ with the S -congruence subgroups form a fundamental system of neighborhoods for $\mathcal{T}$. For Γ an S -congruence subgroup defined by a compact open subgroup $K_1 \subset qK$, one moreover has

$$(2) \quad X \simeq XK/K \simeq q^{-1}K_1/(K \cap q^{-1}K_1) \simeq \Gamma/(\Delta \cap \Gamma) \simeq C_\Delta.$$

Since X is finite, Δ is S -arithmetic. Conversely, Proposition 1 ensures that a fundamental system of S -arithmetic groups is obtained in this way. The filtered ordered set L' is identified with the set of germs, for the topology of S -congruence subgroups, of S -arithmetic groups, and the isomorphism of Proposition 2 is defined as the inverse limit of the isomorphisms (2).

3. Quasi-split groups. Let μ be the group of roots of unity of F . For each non-complex place v of F , let μ_v be the group of roots of unity in F_v ; for a complex v , put $\mu_v = \{1\}$. Deodhar proved that, for G quasi-split, there exists a universal topological central extension $\tilde{G}_v$ of G_v by a discrete group, the “universal covering”, and that for v not real its kernel $\pi_1(G_v)$ is canonically a quotient of μ_v . I have completed this result by showing that in fact $\pi_1(G_v) = \mu_v$. For v real, one has $\pi_1(G_v) = \mathbf{Z}$ or $\mathbf{Z}/2$. In all cases, μ_v occurs as a quotient of $\pi_1(G_v)$.

For almost all v , the extension $\tilde{G}_v$ splits over the natural maximal compact subgroup of G_v . This makes it possible to define the restricted product of the $\tilde{G}_v$, an extension of $G(\mathbf{A})$ by the direct sum of the $\pi_1(G_v)$. Push it out by

$$R : \bigoplus \pi_1(G_v) \rightarrow \bigoplus \mu_v \hookrightarrow \mu, \quad \sigma((x_v)) = \prod_v x_v^{[\mu_v : \mu]}.$$

One obtains a central extension of $G(\mathbf{A})$ by μ , and it follows from Deodhar ⁽²⁾ and C. Moore ⁽³⁾ that this is the universal topological central extension of $G(\mathbf{A})$ by a discrete group which splits over $G(F)$:

$$(3) \quad \begin{array}{ccccccc} 1 & \longrightarrow & \mu & \longrightarrow & \widetilde{G(\mathbf{A})} & \longrightarrow & G(\mathbf{A}) \longrightarrow 1 \\ & & & & \searrow & \uparrow & \\ & & & & & G(F) & \end{array}$$

Take $\widetilde{G}_S$ to be the universal covering of G_S , and write R_S for the restriction of R to

$$\pi_1 G_S = \prod_{v \in S} \pi_1 G_v.$$

Proposition 2 identifies the completion $\widetilde{G(F)}^\wedge(a)$ of $\widetilde{G(F)}$ with the inverse image $\widetilde{G(\mathbf{A})}_S$ of $G_S = \widetilde{G(F)}^\wedge(c)$ in (3), and gives an exact commutative diagram

$$(4) \quad \begin{array}{ccccccc} 1 & \longrightarrow & \mu & \longrightarrow & \widetilde{G(\mathbf{A})}_S & \longrightarrow & G_S \longrightarrow 1 \\ & & \uparrow -R_S & & \uparrow i & & \uparrow \\ \prod_{v \in S} \pi_1(G_v) & \longrightarrow & \widetilde{G(F)}^\wedge(a) & \longrightarrow & G(F)^\wedge(a) & \longrightarrow & 1. \end{array}$$

In particular:

Theorem 1. *Let G/F be quasi-split, and let $\widetilde{G(F)}$ be the inverse image of $G(F)$ in the universal covering of G_S . If*

$$\sum_{v \in S} \text{rank}(G_v) \geq 2,$$

then the intersection of the S -arithmetic subgroups of $\widetilde{G(F)}$ is the kernel of

$$R_S : \prod_{v \in S} \pi_1(G_v) \rightarrow \mu.$$

For $\text{Sp}(2n, \mathbf{Q})$ and $S = \infty$, this is the result announced in the introduction. For $\text{SL}(n, K)$, where K is real quadratic and $S = \infty$, the result obtained generalizes that of J. Millson ⁽⁴⁾.

Problem 1. Does every discrete subgroup of finite covolume in the universal covering of $\text{Sp}(2n, \mathbf{R})$, with $n \geq 2$, contain $2\pi_1 \text{Sp}(2n, \mathbf{R})$? ⁽⁵⁾.

4. The factor system of theta functions. When condition (i) fails, Proposition 1 is no longer available. It is nevertheless still possible to study those S -arithmetic subgroups satisfying a condition of type (1a).

Let S be a nonempty finite set of places, containing all archimedean places, and let $T \supset S$ be such that (G, T) satisfies condition (ii). The interesting case is that in which S consists of a single place at which G has rank 1. Let $\widetilde{G}_S$ again be a topological central extension of G_S by a discrete group, and let $\widetilde{G(F)}$ be the inverse image of $G(F)$ in $\widetilde{G}_S$. For an S -arithmetic subgroup Δ of $\widetilde{G(F)}$, consider the condition:

($\star$) there exists a T -arithmetic subgroup U of $\widetilde{G(F)}$ such that, for every conjugate Δ^g of Δ with $g \in U$, there exists an S -congruence subgroup Γ with

$$\Gamma \cap \Delta = \Gamma \cap \Delta^g$$

(U -invariance of the germ of Δ).

Proposition 3. *The completion $\widetilde{G(F)}^\wedge(a^*)$ of $\widetilde{G(F)}$ for the topology of S -arithmetic subgroups satisfying ($\star$) is a central extension of $\widetilde{G(F)}^\wedge(c) = G_S$.*

More concretely, we have to verify that if (U, Δ) satisfies ($\star$), then Δ satisfies (1c). To prove this, we may first shrink U and Δ . In particular, we may suppose $\Delta \subset U$ by replacing Δ with $\Delta \cap U$. Then U acts by conjugation on the finite group

$$C_\Delta^U = \varprojlim (\Gamma \cap U) / (\Gamma \cap \Delta)$$

where Γ runs over the S -congruence subgroups. Replacing U by the kernel of this action, and Δ by $\Delta \cap \Gamma$, we may also suppose that U acts trivially on C_Δ^U .

Since U is T -arithmetic, its completion $U(c) \subset G_T$ is of finite index in a subgroup $G_{T-S} \times L$, where $L \subset G_T$ is compact open. Since G_{T-S} has no open subgroup of finite index, $U(c)$ itself is of the form $G_{T-S} \times L$. Condition $(\star)$ implies that the $\Delta \cap \Gamma$, with Γ S -congruence, form a fundamental system of neighborhoods of the identity for a group topology on U . The corresponding completion $U(\Delta)$ is an extension of $U(c)$ by C_Δ^U , hence a central extension of $G_{T-S} \times L$. Replace L by an open subgroup L_1 over which this extension is trivial, and replace U and Δ by the inverse images of L_1 . Since G_{T-S} is perfect, a central extension of $G_{T-S} \times L$ is determined by its restrictions to the two factors. This reduces us to the case in which the central extension $U(\Delta)$ of $G_{T-S} \times L$ is the inverse image of a central extension $\tilde{G}_{T-S}$ of G_{T-S} by C_Δ^U . By construction, U lifts into $\tilde{G}_{T-S}$ and Δ is the inverse image of a compact open subgroup K of $\tilde{G}_{T-S} \times L$:

$$(5) \quad \begin{array}{ccc} K & \hookrightarrow & \tilde{G}_{T-S} \times L \\ \uparrow & & \uparrow \\ \Delta & \longrightarrow & U \end{array}$$

Let $\widetilde{G(F)_T}$ be the central extension of $G(F)$ obtained as the inverse image of $G(F)$ in

$$\tilde{G}_T = \tilde{G}_S \times \tilde{G}_{T-S},$$

and let $\tilde{U}$ be the lifting (5) of U into $\widetilde{G(F)_T}$. For $g \in G(F)$, Proposition 1 gives $(g, \Gamma_T) \subset \tilde{U}$ for a T -congruence subgroup Γ_T defined by a sufficiently small $L' \subset L$. If the S -congruence subgroup Γ defined by $K' \subset G_{T-S} \times L'$, with inverse image K'' in $\tilde{G}_{T-S} \times L'$, is small enough that $(g, K') \subset K$, then $(g, \Gamma) \subset \Delta$, which proves (1c).

The kernel

$$\tilde{C}^{\sim*}(S, G) = \text{Ker}(\widetilde{G(F)}^\wedge(a^*) \rightarrow G(F)^\wedge(c))$$

is still described by Proposition 2, with the same proof, and the calculation can be carried out in the quasi-split case, as in no. 3. Here is an application, with $G = \text{SL}(2, \mathbf{Q})$, $S = \{\infty\}$, and $T = \{\infty, p\}$. Let $\widetilde{\text{SL}(2, \mathbf{R})}$ be the double cover of $\text{SL}(2, \mathbf{R})$. The factor systems of theta functions, restricted to arbitrarily small congruence subgroups of $\text{SL}(2, \mathbf{Z})$, may be interpreted as defining a germ τ , for the congruence subgroup topology, of liftings of $\text{SL}(2, \mathbf{Z})$ into $\widetilde{\text{SL}(2, \mathbf{R})}$. This germ is invariant under $\text{GL}(2, \mathbf{Q})$. If one defines a modular form of half-integral weight $k/2$ to be a function on the Poincare upper half-plane which, under the action of a congruence subgroup of $\text{SL}(2, \mathbf{Z})$, transforms like the theta function of a quadratic form in k variables, this amounts to saying that if

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}(2, \mathbf{Q})$$

has positive determinant and f is modular of weight $k/2$, then $f(-z^-)$ and

$$(cz + d)^{k/2} f\left(\frac{az + b}{cz + d}\right)$$

are also modular. Conversely, one has:

Proposition 4. *For every prime p , the germ τ is the unique germ of a lifting of $\text{SL}(2, \mathbf{Z})$ into $\widetilde{\text{SL}(2, \mathbf{R})}$ which is invariant under a subgroup of finite index of $\text{SL}(2, \mathbf{Z}[1/p])$.*

The arguments leading to the theorem of no. 3 show that every arithmetic subgroup of $\text{SL}(2, \mathbf{Z})$ whose germ is invariant under a subgroup of finite index of $\text{SL}(2, \mathbf{Z}[1/p])$ is congruence.

Let $\Gamma \subset \text{SL}(2, \mathbf{Z})$ be a congruence subgroup, and let τ_1 and τ_2 be two liftings of Γ into $\widetilde{\text{SL}(2, \mathbf{R})}$. Let Δ be the subgroup of Γ on which $\tau_1 = \tau_2$. If the germs of τ_1 and τ_2 are distinct and invariant under a $\{p\}$ -arithmetic group, then Δ satisfies $(\star)$ and is not a congruence subgroup.

(*) Session of 3 July 1978.

(¹) M. S. RAGHUNATHAN, *Publ. Math. I.H.E.S.* 46 (1976), 107–162.

- (²) V. V. DEODHAR, *On central extensions of rational points of algebraic groups* (to appear).
- (³) C. C. MOORE, *Publ. Math. I.H.E.S.* 35 (1968), 5–70.
- (⁴) J. MILLSON, *Real Vector Bundles with Discrete Structure Groups* (to appear).
- (⁵) I learn from Mostow that such a subgroup Γ at least contains a subgroup of finite index in $\pi_1 \mathrm{Sp}(2n, \mathbf{R})$: one checks that the projection of Γ into $\mathrm{Sp}(2n, \mathbf{R})$ is discrete; otherwise the Lie algebra of its closure would be nontrivially central.

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